SLEEVEDRIFT · DESIGN DECISION RECORD

SleeveDrift design for construction

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/decisions/0002-design-for-construction.md @ 1ba9c34		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Constructability review and the changes that make the design buildable, decided under Amish's pre-approvals of 2026-10-03

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted

Context

STANDARDS section 18 requires a constructable design at TRL 3: every part made by a stated process, and every part fitting and fastened to its neighbours. The concept (SVD-PRC-001 v0.2) showed a one-piece hooded sleeve, a return pulley "fixed to the ring", an endless loop with sledges, a hand capstan and an air duct, without saying how any of them is made, carried into a pipe or fixed. The constructability review went component by component, then checked the build123d model for overlaps between every pair of components and for parts that touch nothing (`python cad/src/model.py --check`: no overlaps and nothing floating). Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible". The changes below were decided under his pre-approvals of 2026-10-03 ("I pre-approve the batch runs along with any recommendations you come up with." and "Proceed with the remaining 15 scaffolds").

> **Safety:** Changes 1, 2, 4, 7 and 8 carry loads that protect people in the pipe; each is load tested before use (SVD-BLD-001, section 6).

Options considered

For each part: make it as the concept drew it, or change it to the simplest physically sound form that keeps what it does. Changes that would alter what SleeveDrift does, its pitch or its safety case were not in scope.

Decision

Table 1. Changes made for construction.

#	Component	The concept had	The constructable design has	Why
1	Hooded cutting ring	One short sleeve	Four rolled 5 mm S355 segments (crown 120 degrees with the hood, two sides of 90 degrees, bottom of 60 degrees), bolted through 50 x 10 mm flanges inside with four M12 bolts per joint	A one-piece ring is about 85 kg and cannot be carried along a pipe; the heaviest segment is 31 kg. No joint lies on the floor lane
2	Ring fixing	Not defined	Ring rests on the pipe invert; four M20 jacking screws in weld nuts (15 degrees either side of the top and 80 degrees) press swivel pads on the pipe wall	No drilling, welding or power at the face; the hood load goes into the pipe through contact, without free play
3	Hood	A hood over the top third	The crown runs 250 mm beyond the ring front; its flanges run the full 700 mm as hood ribs; a 60 x 10 mm lip inside the front edge; cutting bands with a 30 degree bevel on the other segments	Stiffness and an edge that can meet debris
4	Return pulley	Fixed to the ring	A bought 240 mm sheave lying flat on a 25 mm pin, on a 10 mm base plate with two saddles cut to the shell, welded to the bottom segment; spacer, keeper and R-clip; a 4 mm chequer-plate cover on four posts	The sheave needs a flat, square seat inside a curved shell; the cover guards it and is a kneeling plate
5	Rope legs	Along the floor	Train lane on the centre line; return leg on the floor 230 mm to the +Y side; both rise to the sheave plane, 185 mm up, over the last 1.2 m, so the train stops 1.2 m short of the mouth	Keeps the ropes clear of the ring rear and the jacking screws
6	Sledges	Low curved trays	Folded 1.5 mm galvanised trays, 160 mm base flaring to 300 mm, 130 mm deep, 1 m long (29.9 L), on two UHMW-PE runners that sit on the pipe curve; 8 mm end lugs	A form any sheet metal shop can fold; the runners set the friction
7	Loop	Endless rope with clip-on sledges	A shuttle train of four sledges joined by connecting links, with a 2.5 kN breakaway swivel at each end between the train and the ropes	See SVD-DDR-001, items 3 and 5
8	Portal haul station	A hand capstan	Two bought two-speed self-tailing winches on two welded 40 x 40 x 3 mm trestles (34.8 kg each) tied by four bolted cross ties; anchor eyes level with the rope; round slings to a structural anchor; station 5.5 m behind the mouth so the rope rises 6.6 degrees to the drums	Each trestle is one carry; the sling at rope height leaves no overturning moment; the lead suits a self-tailer
9	Air outlet	A duct outlet behind the hood	A 194 mm steel spigot in two saddles welded inside the crown at 40 degrees toward +Y; duct hung 265 mm off the pipe axis on magnet hangers every 2.5 m; crown jacking screws placed at 15 degrees to clear it	The layflat needs a rigid end, and the duct must leave a crawl space (554 mm clear circle)
10	Signal and light	Signal line and lamps	Cable on magnet hangers near the crown; magnetic lamp inside the hood	Nothing drilled into the pipe
11				See SVD-DDR-001, item 9

#	Component	The concept had	The constructable design has	Why
	Casualty sledge	Full-length sledge	Bought roll-up stretcher with a made 450 mm spreader bar and rope bridle; the swivels clip to its eye	
12	Set-up	Not defined	Bottom segment first, taken in on a sledge by a crew member with both ropes paying out; the other three segments together on one shuttle trip	The sheave must be at the face before the shuttle can work; about 1.6 h in all

Consequences

- `cad/src/model.py` carries these changes; STEP and STL, drawings SVD-DWG-001 and 002 (Rev P2), the concept media and the build plan pictures are generated from it.
- The calculations in SVD-CAL-001 are on this design.
- `design_state: constructable` in `project.yaml`.
- Items that can only be settled with real parts (winch base pattern and power ratios, sheave bore and rating, swivel release load, pad friction) are listed under "To confirm when parts are bought" in SVD-DEC-001.